

CrewConnect – Interview Introduction

Project Introduction

CrewConnect is a Django-based HR and Employee Management System. We have two major roles: HR and Employee.

Tech Stack: Python, Django, SQLite, HTML, CSS, Bootstrap, JavaScript, Django ORM, Django Authentication, and SMTP for email integration.

HR Module

HR can manage **employees, departments and designations**, approve employee leave requests, generate payslips and view payroll reports.

HR can add, view, update and delete employees.

While creating an employee, HR provides Employee ID, Name, Email, Phone, Department, Designation, Joining Date, Employment Type, Salary, Address and Status.

Departments

We maintain separate Department and Designation master tables.

We maintain different departments based on the organization's requirements, such as **HR, Development, Testing and Finance**.

Designations

We maintain different designations such as **Python Developer, Senior Python Developer, Software Tester, HR Executive and Accountant**.

Employment Types

We have predefined employment types: **Permanent, Contract, Part-time and Intern**. These are implemented using Django model choices.

Employee Module

Employees can **view their own profile, apply for leave, view leave history and leave calendar, and access their payslips**.

Authentication

For authentication, we use **Django's built-in User model**. We maintain a **OneToOne relationship between User and Employee**.

When HR creates an employee, the system creates the Django User account with an **unusable password** and sends the employee an email containing a secure password setup link.

The employee opens the link, creates their password and then logs in.

Login Differentiation

After login, we differentiate the user based on whether an **Employee record exists for that User** and redirect them to the appropriate dashboard.

Leave Management

Employees can apply for different leave types such as **EL – Earned Leave, SL – Sick Leave and CL – Casual Leave**.

When an employee applies for leave, the request initially has a **Pending** status. HR can approve or reject the leave.

Employees can view their leave status and leave calendar.

Payroll

HR generates monthly payslips containing **basic salary, allowances, deductions and net salary**.

Net Salary = Basic Salary + Allowances - Deductions

We added a **database-level unique constraint** to prevent duplicate payslips for the same employee and month/year.

Reports

We have an **HR reporting module** where HR can select a month and year and view aggregated payroll information using Django ORM's aggregation functions such as **Sum**.

Frontend

On the frontend, we use **HTML, CSS and Bootstrap** to create dashboards and responsive UI.

We use Django templates with template inheritance so common components such as the sidebar and layout do not need to be repeated on every page.

Backend

On the backend, Django handles **URL routing, views, models, authentication, sessions and database operations through Django ORM**.

SQLite is currently used as the database, and **SMTP** is used for sending employee onboarding emails.

Overall Project Workflow

Employee Onboarding → Authentication → Leave Management → Leave Approval → Payroll → Payslips → HR Reports

Stage	Flow
1	Employee Onboarding
2	Authentication
3	HR / Employee Dashboard
4	Employee Management / Profile & Leave
5	Leave Approval
6	Payroll & Payslips
7	HR Reports

Final Interview Line

The overall workflow is employee onboarding, authentication, leave management and approval, payroll and payslip generation, followed by HR reporting.